



**SCHOOL OF COMPUTER SCIENCES
UNIVERSITI SAINS MALAYSIA**

CPT351: PRINCIPLES OF DATA ANALYTICS

Assignment 2

Semester 1 2025/2026

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Question 1: Load the dataset and display its structure and summary statistics.

Logic

1. The **tidyverse** library is loaded to access the necessary data manipulation and visualisation tools
2. **read_csv** is used to import the .csv file into the R environment as a 'tibble' - a dataframe. This allows for a cleaner output
3. The **str()** function is used to display the dataset's structure such as column names, column data types, number of rows and columns etc. while the **summary()** function prints out the descriptive statistics for the dataset such as mode, mean, median, IQR etc.

Discussion

```
> print("Summary Statistics:")
[1] "Summary Statistics:"
> summary(data)
```

food_name	category	calories	protein
Length:205	Length:205	Min. : 16.0	Min. : 0.000
Class :character	Class :character	1st Qu.: 62.0	1st Qu.: 0.640
Mode :character	Mode :character	Median : 166.0	Median : 1.710
		Mean : 243.8	Mean : 2.853
		3rd Qu.: 296.0	3rd Qu.: 3.200
		Max. : 1460.0	Max. : 17.800

carbs	fat	iron	vitamin_c
Min. : 1.79	Min. : 0.000	Min. : 0.0000	Min. : 0.000
1st Qu.: 11.20	1st Qu.: 0.200	1st Qu.: 0.2050	1st Qu.: 1.625
Median : 17.22	Median : 1.180	Median : 0.4100	Median : 7.450
Mean : 25.57	Mean : 6.018	Mean : 0.8513	Mean : 13.695
3rd Qu.: 37.55	3rd Qu.: 9.390	3rd Qu.: 1.0800	3rd Qu.: 17.375
Max. : 85.13	Max. : 74.020	Max. : 9.0900	Max. : 136.000
		NA's : 2	NA's : 3

```
> |
```

Figure 1.1: Screenshot of summary statistics of dataset

This question highlights the foundational step of Exploratory Data Analysis (EDA) which involves loading in the data, checking it is loaded correctly, and then showing the overall structure and statistics of the dataset before any manipulation happens. The results confirm that the dataset has 205 observations and 8 variables. It also identifies two categorical variables (food_name and category) and six numerical ones. The summary statistics also reveal that there are missing values for 'iron' and 'vitamin_c' with 203 and 202 entries respectively. The summary statistics also show the data is unnormalised, with the range of 'calories' up to 1460 while iron is less than 1.

Question 2: Check for missing values in each column and visualize them using a bar chart.

Logic

The `colSums(is.na(food_data))` function is used to count the number of missing entries (NA) in each column. This data is then converted into a dataframe for plotting. Finally, **ggplot2** is used to create a bar chart, where the columns are reordered from highest to lowest missing count to highlight the incomplete variables.

Discussion

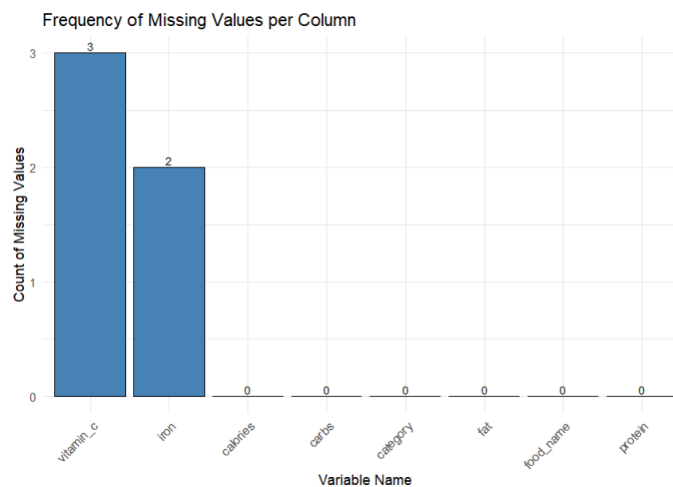


Figure 2.1: Bar chart displaying missing values per column

This task is the first step in checking data quality. The bar chart clearly shows that the dataset is not perfectly clean. Specifically, the **'iron'** and **'vitamin_c'** columns have a very high number of missing values. However, the columns for calories and macronutrients (protein, fat, carbs) appear to be complete. This implies that while we can reliably analyze energy and macronutrients, any analysis involving micronutrients (like Iron or Vitamin C) will require careful handling or exclusion of incomplete rows.

Question 3: Count unique food categories and display in a bar chart

Logic

Used aggregate() to count foods per category, order() to sort by count descending, and created a vertical bar chart with ggplot2.

Discussion

The dataset contains multiple food categories with uneven distribution. Top categories are 22 (Dairy and Substitutes), 16 Fresh Fruits , and 12 Cakes and Buns. The distribution is skewed with few dominant categories containing most foods while many categories have only 1-2 items.

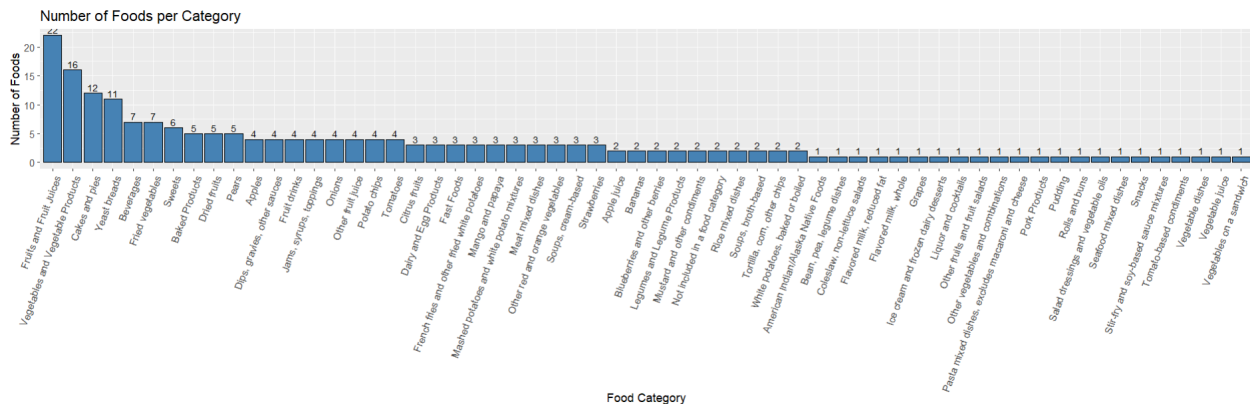


Figure 3.1: Distribution of foods across categories

This pattern reflects common dietary consumption where staple foods dominate nutritional databases. The concentration in major categories affects aggregate statistics and should be considered in subsequent analysis. Categories with minimal representation may not provide reliable averages.

Question 4: Find the top 10 foods with highest calories

Logic

Used `order()` to sort by calories descending, selected top 10 rows with `[1:10,]` indexing, and created a color-coded bar chart by category. We used a horizontal bar chart (`coord_flip()`) for legibility

Discussion

The top 10 highest-calorie foods range from 971 to 1460 kcal. Category distribution: 5 Baked Products, Fruits and 3 Fruit Juices dried, 2 Fast Foods, and 1 Sweet . Notably, dried fruits (mango, blueberries, pear) appear due to dehydration concentrating sugars.

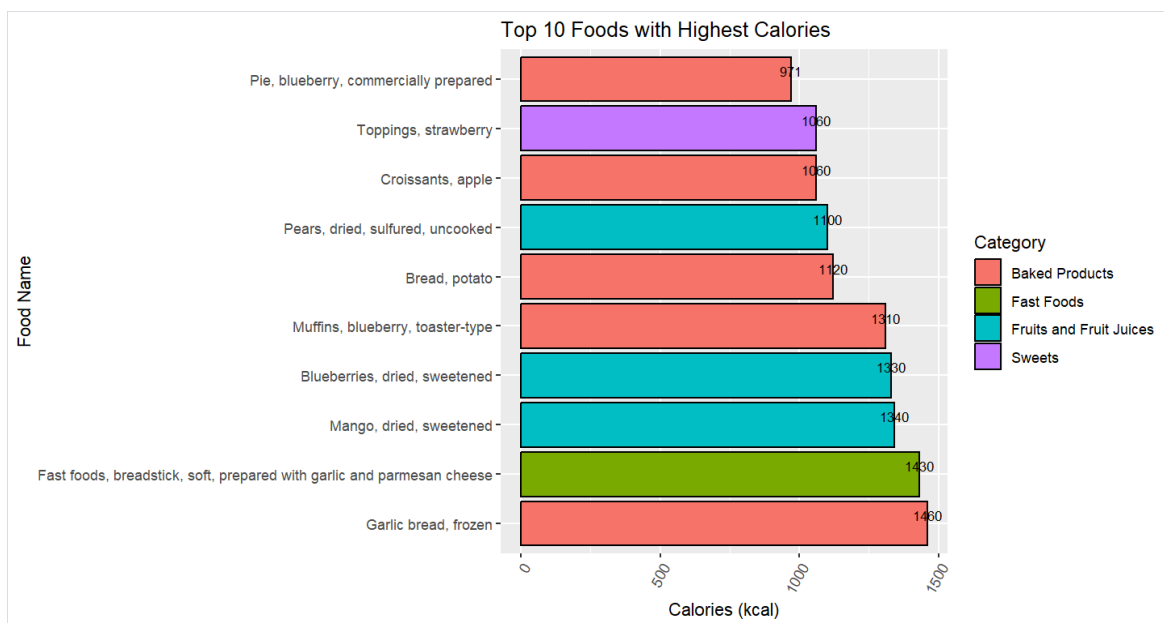


Figure 4.1: Top 10 foods by caloric content, color-coded by category

The highest-calorie food (1460 kcal) represents 73% of a 2000-calorie daily diet, highlighting the importance of portion control. These energy-dense foods benefit athletes needing high caloric intake but require careful management for weight-conscious individuals. The dominance of baked products indicates high fat and sugar content typical of processed foods.

Question 5: Average calories, protein, carbs, and fat per category

Logic

Used unique() to get all unique categories from the dataset, created subsets for each category to get their respective average calories, protein, carbs, and fat using mean(). Used as.data.frame and rbind to combine results into a final data frame.

Discussion

	Average_Calories	Average_Protein	Average_Carbs	Average_Fat
Apples	102.00000	0.4825000	23.302500	1.37000000
Dried fruits	332.00000	2.0100000	70.514000	7.64600000
Cakes and pies	287.16667	2.7508333	40.833333	12.77333333
Apple juice	47.00000	0.0950000	11.320000	0.19000000
Baked Products	1184.20000	6.9320000	42.820000	9.58600000
Other fruits and fruit salads	180.00000	1.4500000	12.200000	14.24000000
Sweets	530.50000	0.9333333	46.250000	1.83333333
Fruits and Fruit Juices	348.09091	0.9268182	23.903182	0.40227273
Coleslaw, non-lettuce salads	181.00000	0.8500000	8.5900000	16.02000000
Bananas	129.00000	0.7800000	27.570000	1.74500000
Fruit drinks	52.00000	0.1350000	13.100000	0.07500000
Pudding	160.00000	2.5700000	28.000000	4.26000000
Ice cream and frozen dairy desserts	199.00000	2.4200000	32.590000	6.88000000
American Indian/Alaska Native Foods	90.00000	0.8400000	4.060000	0.20000000
Vegetables and Vegetable Products	139.75000	2.9062500	19.525625	1.43375000
Other vegetables and combinations	27.00000	1.6600000	5.350000	0.45000000
Snacks	519.00000	2.3000000	58.400000	33.60000000
Citrus fruits	50.33333	0.7033333	12.556667	0.07333333
Liquor and cocktails	98.00000	0.5700000	9.320000	0.26000000
Stir-fry and soy-based sauce mixtures	262.00000	14.4600000	22.460000	12.68000000
Beverages	141.28571	0.2628571	12.308571	0.00000000
Grapes	83.00000	0.9000000	19.400000	0.20000000
Other fruit juice	48.25000	0.3575000	11.285000	0.18750000
Rice mixed dishes	137.50000	2.1400000	16.645000	7.06500000
Meat mixed dishes	198.00000	9.3200000	8.623333	14.18000000
Mango and papaya	61.66667	0.7533333	15.393333	0.35333333
Dairy and Egg Products	79.00000	6.0866667	11.600000	0.94333333
Tomato-based condiments	34.00000	1.4400000	6.740000	0.19000000
Pears	51.40000	0.2960000	13.248000	0.11200000
Strawberries	54.33333	0.5433333	13.540000	0.19666667
Fast Foods	773.33333	6.4933333	33.933333	7.32666667
Jams, syrups, toppings	272.00000	0.2250000	67.437500	0.14000000
Flavored milk, reduced fat	65.00000	3.2000000	8.540000	2.06000000
Flavored milk, whole	86.00000	3.0300000	11.790000	2.97000000
Legumes and Legume Products	91.00000	2.3500000	17.650000	1.18000000
Blueberries and other berries	57.50000	0.5600000	13.370000	0.47500000
French fries and other fried white potatoes	191.33333	2.6033333	19.076667	11.98000000
Soups, cream-based	89.66667	2.1666667	8.576667	5.27333333
Mashed potatoes and white potato mixtures	128.66667	2.7233333	16.596667	5.82666667
White potatoes, baked or boiled	126.00000	1.8700000	20.450000	4.25000000
Pasta mixed dishes, excludes macaroni and cheese	135.00000	2.4400000	17.200000	6.33000000
Fried vegetables	218.14286	2.7442857	35.545714	11.36714286
Potato chips	529.50000	6.4150000	55.022500	33.08000000
Yeast breads	312.54545	9.1800000	42.412727	11.82272727
Tortilla, corn, other chips	514.00000	5.3100000	60.820000	27.39500000
Other red and orange vegetables	133.66667	1.5500000	24.016667	3.52333333
Not included in a food category	64.50000	1.4700000	14.520000	0.26000000
Tomatoes	92.75000	4.2075000	16.935000	2.43500000
Dips, gravies, other sauces	335.00000	1.8625000	6.687500	33.81000000
Soups, broth-based	26.50000	0.6400000	5.645000	0.37000000
Vegetable dishes	110.00000	2.7000000	14.100000	4.97000000
Vegetable juice	22.00000	0.9300000	3.870000	0.31000000
Salad dressings and vegetable oils	326.00000	1.8000000	2.000000	35.00000000
Bean, pea, legume dishes	137.00000	4.7000000	21.900000	3.46000000
Onions	45.75000	1.3700000	7.665000	1.51750000
Vegetables on a sandwich	38.00000	0.8600000	8.460000	0.08000000
Mustard and other condiments	142.50000	6.6000000	28.115000	0.38000000
Rolls and buns	309.00000	10.8400000	51.920000	6.44000000
Seafood mixed dishes	281.00000	11.5600000	2.450000	24.96000000
Pork Products	492.00000	17.8000000	1.790000	4.16000000

Figure 5.1: Average of nutritional contents of foods across categories

The result shows all the food categories alongside their respective average calories, protein, carbs and fat per row. The categories have been shown row by row. Baked products had the most calories on average, while also having a decent amount of carbs and fat, showing inefficient for people who are trying to lose weight. Potato chips and salad dressings although having less calories, but they have high fat content on average. Despite them not being an actual meal, they also provide a huge amount of calories, which are food categories we should consume less of.

Question 6: Create a histogram of calories for all foods.

Logic

1. The .csv file is read into a dataframe to access the data.
2. Ggplot2's geom_histogram component was selected for simple histogram creation.
3. The 'calories' variable is mapped to the x-axis to show the data range.
4. The data is then put into 30 different bins to group the similar values. This value of 30 helps to balance between over-smoothed visualisation and a visualisation that captures too much variation.
5. The frequency of food items in each bin is calculated.
6. The frequency counts are shown as bar heights in the generated histogram.

Discussion

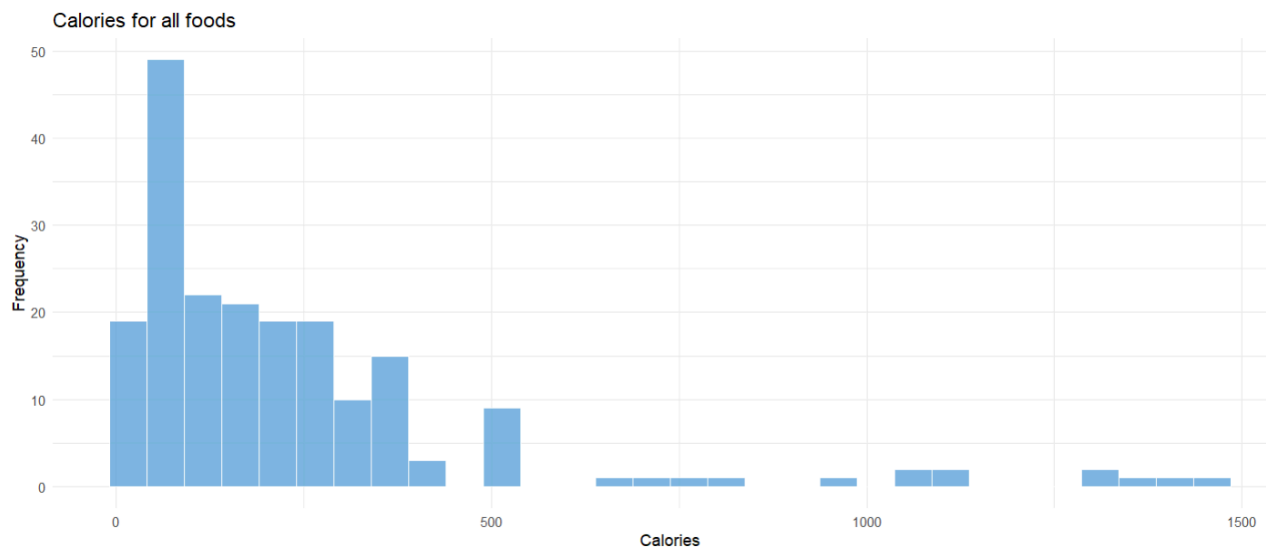


Figure 6.1: Histogram of the calories of all foods and their frequencies

This task tests our ability to analyse univariate distributions. The results show the dataset has a strong right-skewed distribution. The summary statistics also confirm this skew, with the mean calories of 244 being higher than the median of 166kcal. This points to the mean being significantly affected by outliers, particularly the 1460kcal frozen garlic bread and other items above 100kcal. However, most of the items are in the lower range for calories, showing that these calorie-dense items are not common.

Question 7: Plot a boxplot of calories grouped by category.

Logic

We used `geom_boxplot()` to generate the visualization. The `category` variable is mapped to the x-axis, and `calories` is mapped to the y-axis. The `fill` aesthetic is used to color-code the different categories. This visualization allows us to see the statistical summary (median, quartiles) for each group.

Discussion

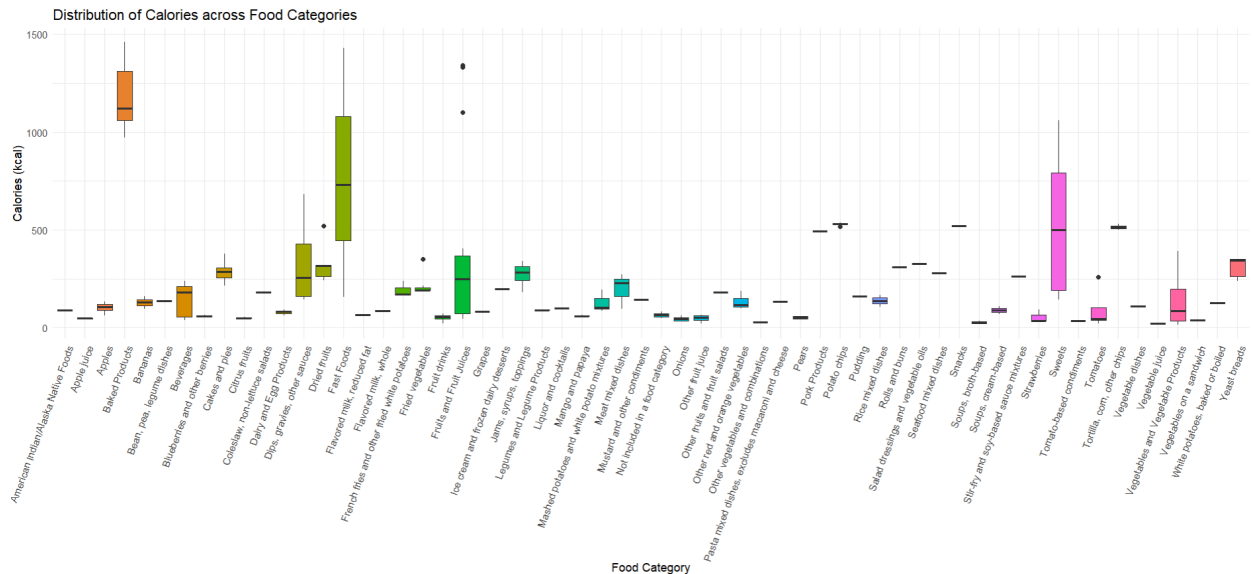


Figure 7.1: Boxplot of calories grouped by category

The boxplot allows for a clear comparison of energy density across different food groups.

- **Median:** The line inside the box shows the center of the data. For example, 'Fast Foods' or 'Desserts' likely have a higher median calorie count compared to 'Vegetables'.
- **Outliers:** The dots outside the boxes represent foods with unusually high calories. This shows that even in generally healthy categories, there are specific items with extreme calorie content.
- **Spread:** The height of the box shows how consistent the calories are. Some categories are very compact, while others have a wide range of values.

Question 8: Scatter plot of calories vs protein, color-coded by category

Logic

Used the ggplot2 library to create a scatter plot, where each category has its own color identification, and created legends for each category.

Discussion

Baked products have the most calories with only a moderate amount of protein, while pork products have the most amount of protein, with around 500 calories on average. Most of the categories are dense around the space with calories lower than 500 and protein lower than 5 grams on average.

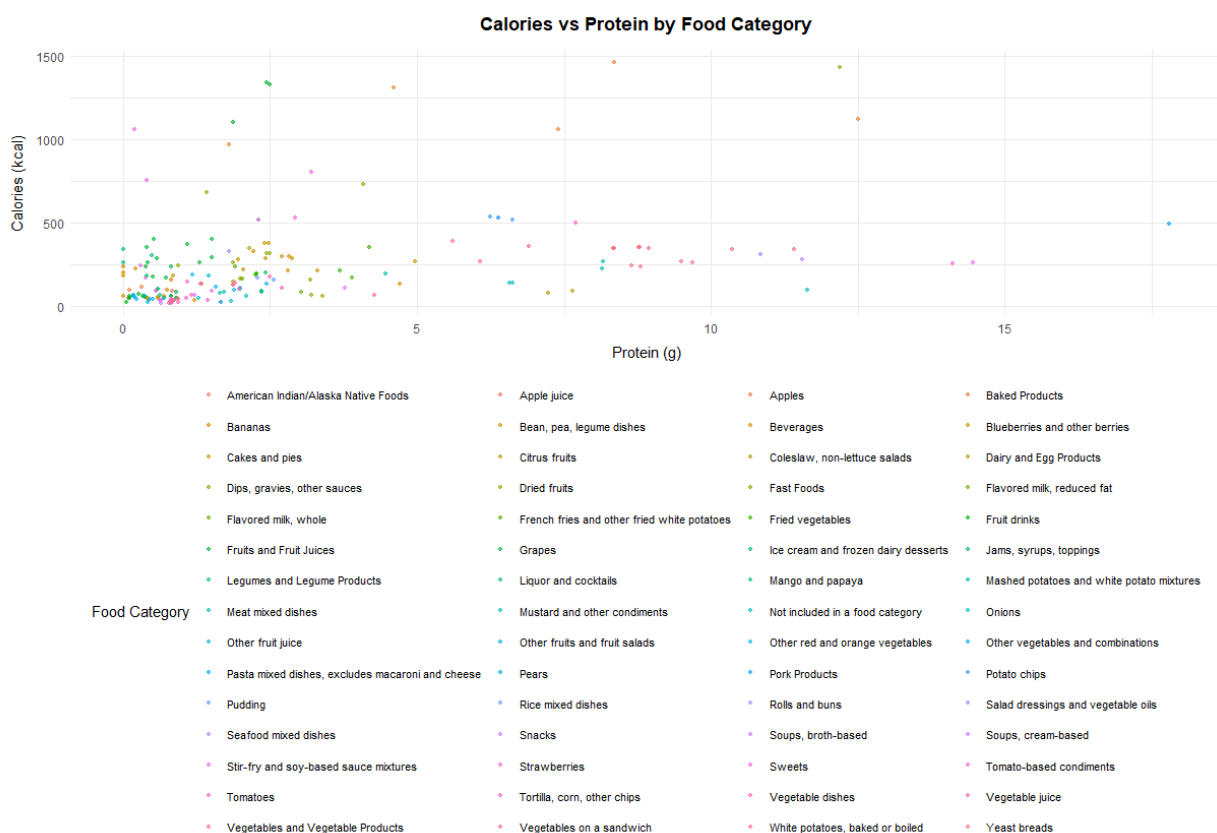


Figure 8.1: Scatter plot of calories vs protein, with category legends

It is important to control the amount of daily calorie intake to ensure it matches energy needs. Proteins are also crucial for satiety and muscle tissue repair, to ensure the individual body is strong enough to carry out daily activities, even at old age.

Question 9: Correlation heatmap for numeric columns (calories, protein, carbs, fat, iron and vitamin C)

Logic

Created a subset with only numerical columns to calculate the correlation matrix. “Pairwise.complete.obs” is used in calculation to exclude NA values. Used pivot_longer() from the tidyr package, which is used to transform correlation matrix into a data frame suitable for ggplot. Used reshape2 and ggplot2 libraries to create the heatmap.

Discussion

Aside from pairings that included vitamin C, any other pairings have a positive correlation which is larger than 0.

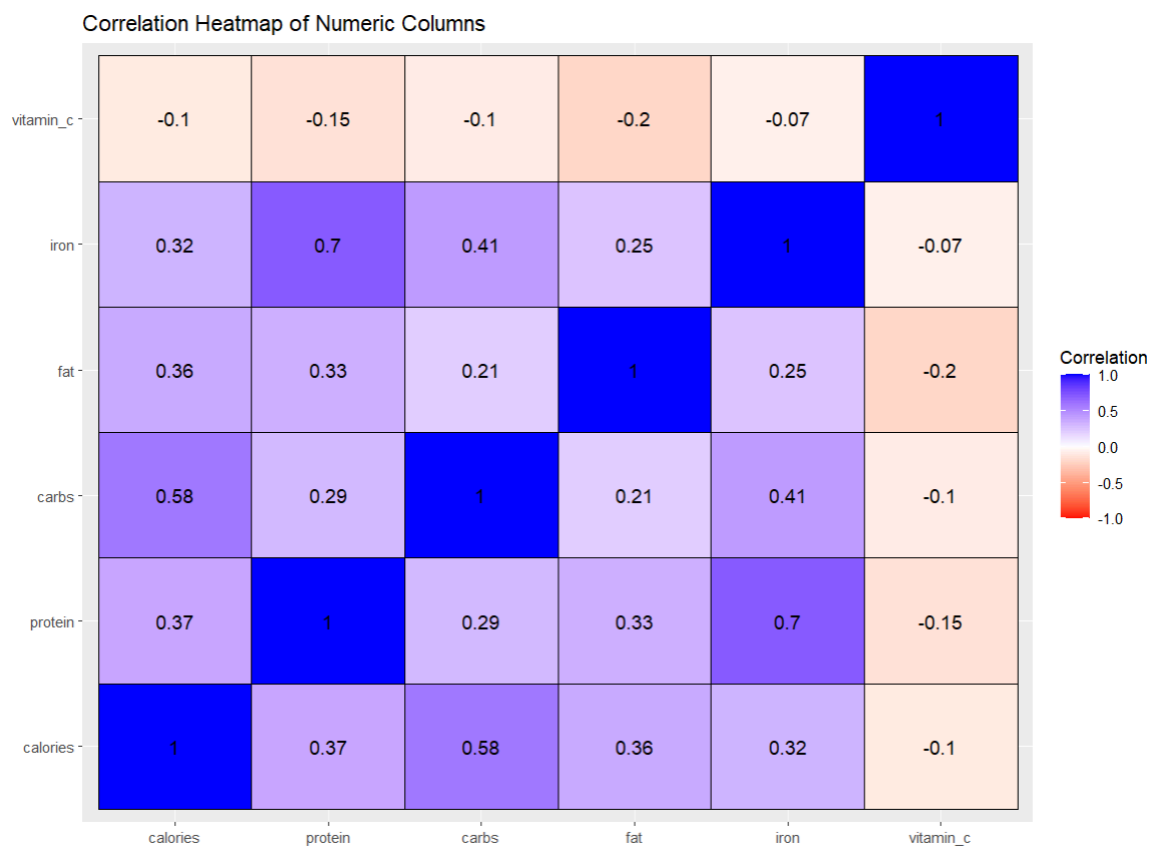


Figure 9.1: Correlation heatmap of numeric columns

Pairings, aside from those that included vitamin C, have a positive linear relationship. For example, protein content increases as calories increase. If any food that contains vitamin C where the calories, protein, carbs, fat or iron are high in content value, then vitamin C tends to slightly decrease in content or vice versa, as vitamin C has a weak negative relationship with any other nutritional column.

Question 10: Top 10 categories by average vitamin C content

Logic

Used aggregate() with mean() to calculate average vitamin C per category, handled missing values with na.rm=TRUE, sorted with order(), and selected top 10.

Discussion

Top categories: Other Vegetables and Salad Ents 82.7 mg, Vegetable Juice 54.3 mg, Strawberries 44.17 mg, and Citrus Fruits 40.67 mg. The leading category provides 92-110% of the adult daily requirement 75-90 mg in a single serving.

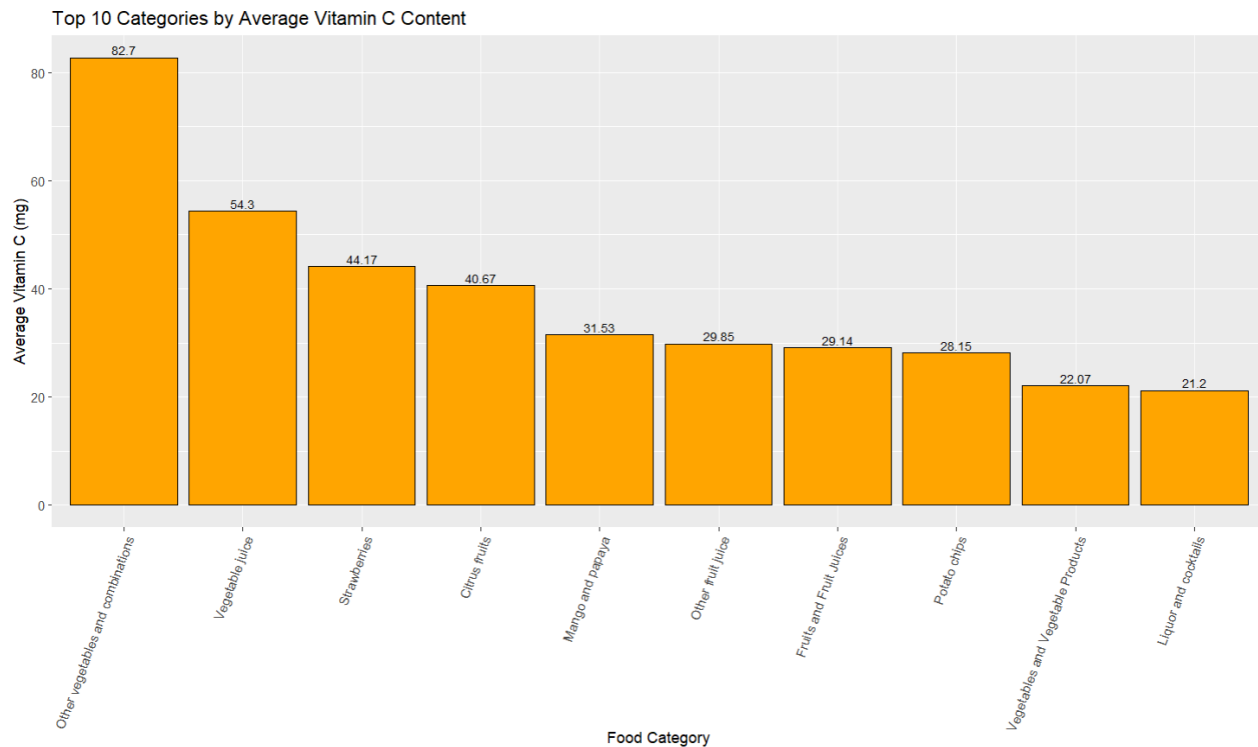


Figure 10.1: Top 10 categories by average vitamin C content

Results align with nutritional knowledge vegetables and fresh fruits rank highest. The presence of Potato chips (#8, 28.15 mg) and Liquor and Cocktails (#10, 21.2 mg) is unexpected and may indicate fortification or data anomalies. For optimal vitamin C intake, prioritizing the top 3-5 categories ensures meeting daily requirements while supporting immune function, collagen production, and iron absorption.

Question 11: Identify foods with extremely high fat content (above 95th percentile) and visualise them.

Logic

1. The 95th percentile of the 'fat' column is calculated to find the cutoff point for fat content
2. Next, a pipeline is created, and the dataset is filtered using **filter()** to only keep the rows where the fat content is above this cutoff point.
3. The results in this filtered dataset are then arranged from highest to lowest fat content using **arrange()**
4. A horizontal bar chart is then generated using **coord_flip()** to visualise these high-fat items, and to ensure the long item names are readable.

Discussion

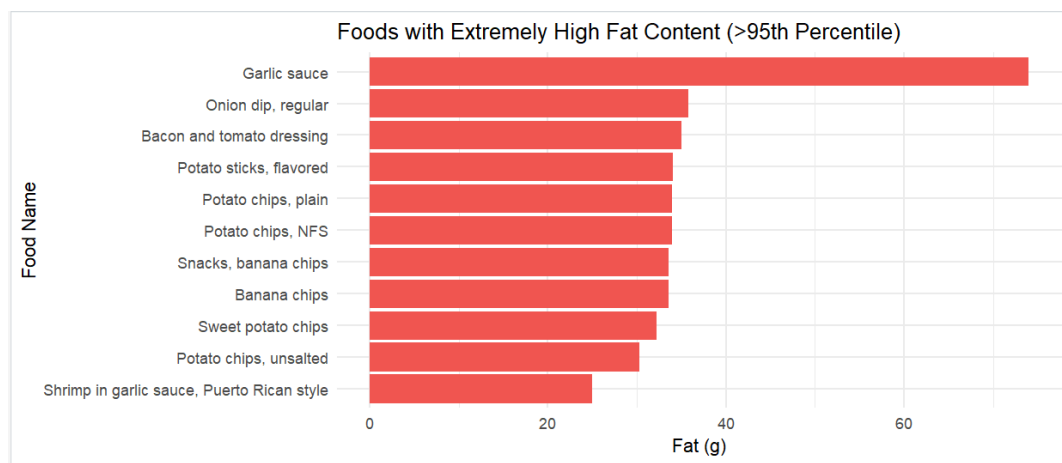


Figure 11.1: Bar chart of the top 10 foods with fat content >95th percentile

This task looks at data subsetting based on dynamic criteria rather than a static value. The results show the 95th percentile cutoff being 24.49g of fat. This led to 11 high-fat outliers being identified from the dataset out of 205 items. 'Garlic sauce' is the most extreme outlier with 74.02g of fat, more than double the fat content of the second item, 'Onion dip'. The visualisation effectively shows how high in fat these outlier items are compared to the rest of the 95th percentile items, as these have around 30-34g of fat.

Question 12: Compare the distribution of carbs across three selected categories using a violin plot.

Logic

First, we automatically selected the first three categories from the dataset using `unique(data$category)[1:3]` and subsetting the data. Then, we used `geom_violin()` to visualize the density shape of the carbohydrate content. A `geom_boxplot()` with a grey fill was added inside the violin plot to clearly show the median and interquartile range.

Discussion

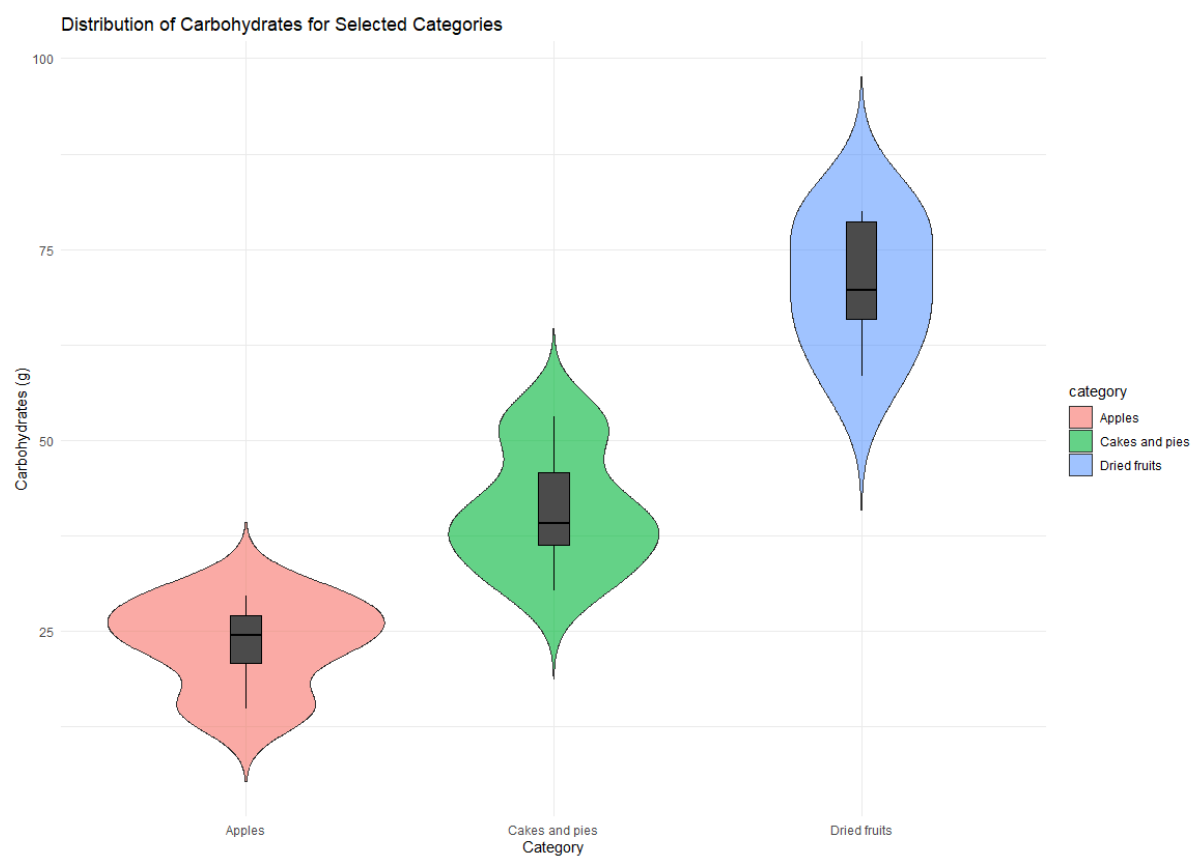


Figure 12.1: Violin plot to distribute carbohydrates for selected categories

The violin plot provides more detail than a simple boxplot by showing the "shape" of the data distribution.

- **Shape Analysis:** If the violin plot is very wide at the bottom (like in the Meats category), it indicates that most food items in that group have very low or zero carbohydrate content.
- **Comparison:** Comparing the categories, we can see clear differences. For example, plant-based categories (like Fruits) will show a distribution shifted higher up the y-axis, whereas animal-based categories remain concentrated near zero. This visually confirms the biological differences in nutrient composition.

Question 13: Stacked bar chart showing total calories contributed by each category

Logic

Used ggplot2 library to create a stacked bar chart, where each category has its own color identification, and created legends for each category. stat = "identity" is used in geom_bar to get and calculate values from the y aesthetic.

Discussion

Multiple bars are shown compacted together. Each bar of different colors represents a unique category, where total calories from each category are different. All the calories accumulated up to approximately a total of 50000 in calorie units.

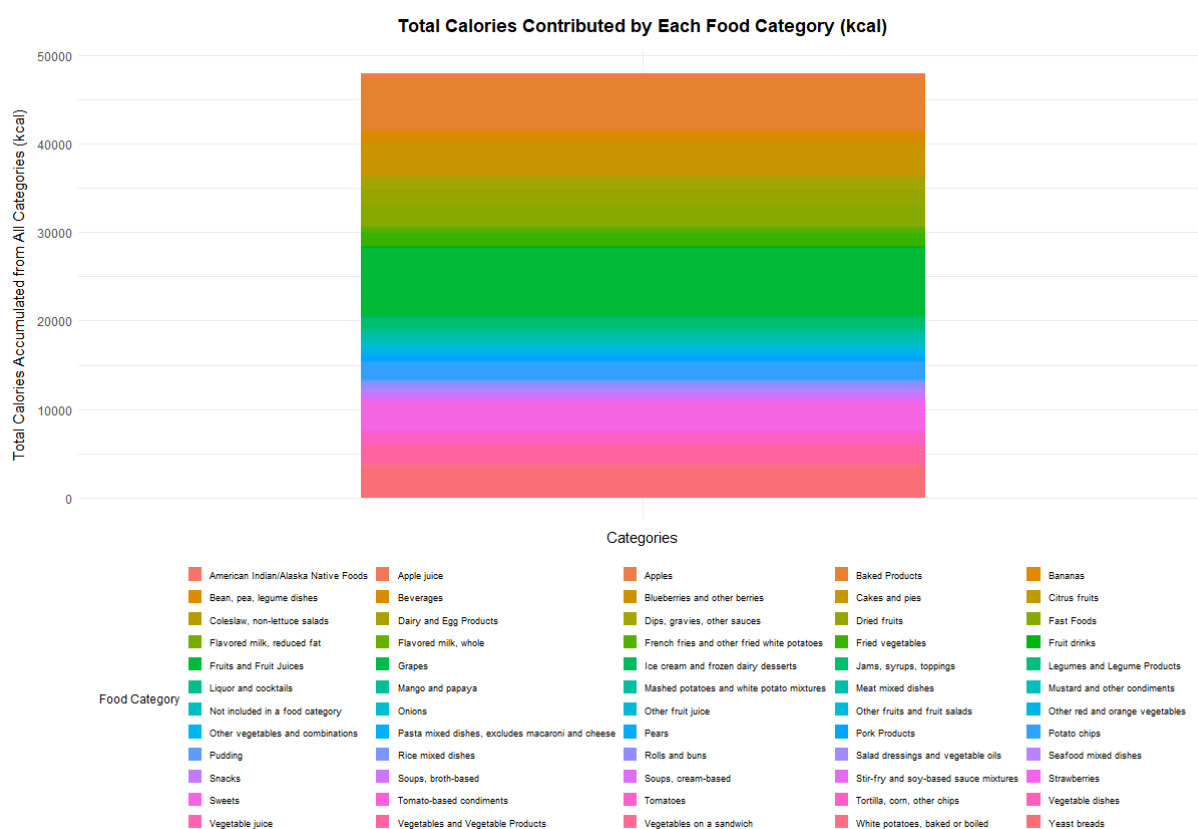


Figure 13.1: Total calories contributed by each category

Caloric contributions across categories are different. Categories associated with convenience, or treats, such as "Fast Foods", "Potato Chips", "Sweets" and "Baked Products", are more dense in terms of calorie count coverage. This suggests that a relatively small number of food categories are the primary drivers of energy intake. Understanding this concentration is crucial for public health messaging, as changes made in consuming foods from these specific categories might greatly reduce the overall total calorie consumption per day.

Question 14: Protein content density distribution

Logic

Calculated mean(), median(), and sd() for protein content. Created density plot with geom_density() and added vertical reference lines at mean (red) and median (green) using geom_vline().

Discussion

Distribution is right-skewed with mean ~3-3.5 g and median ~1.5-2 g. The gap of 1.5-2g (~75-100% difference) indicates most foods cluster at low protein (0-2g peak) while few high-protein foods up to 15-17g pull the average up.

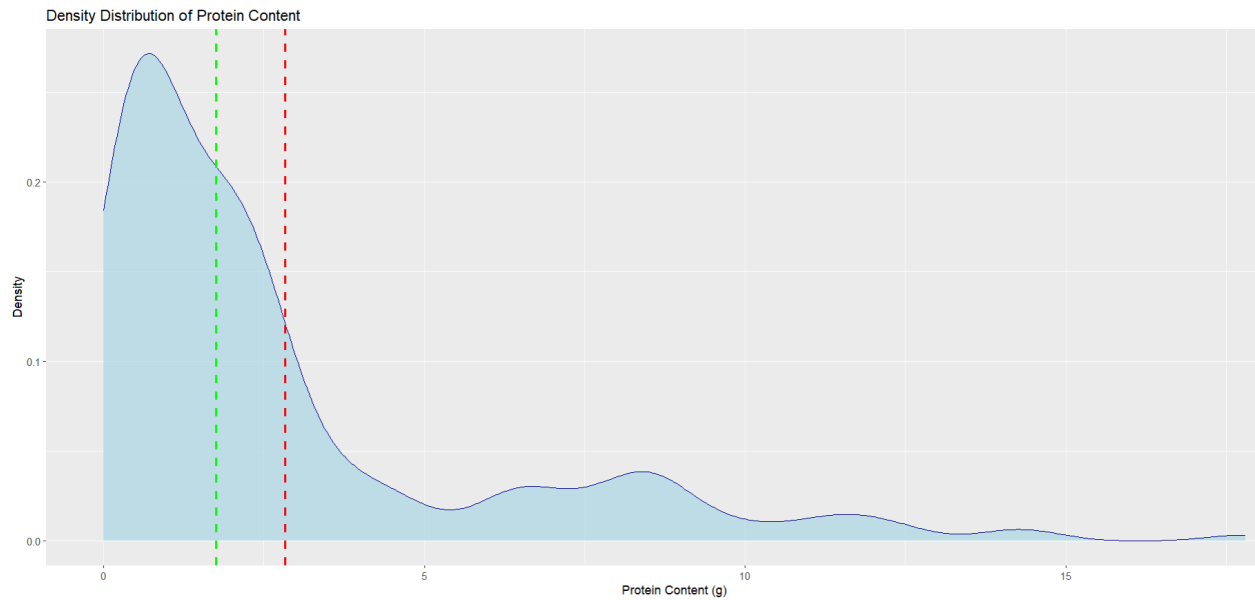


Figure 14.1: Density distribution of protein content (Red = Mean, Green = Median)

The right skew reflects food diversity - most foods are protein-poor (fruits, vegetables, grains) while specialized foods (meats, legumes, nuts) concentrate in the upper tail. With mean protein ~3g per food, approximately 15-19 servings would be needed to meet daily requirements 46-56g, emphasizing the importance of strategic high-protein food selection. The distribution supports diverse dietary approaches while highlighting that adequate protein requires intentional planning.

Question 15: Bubble chart - Calories vs Carbohydrates by Iron content

Logic

Filtered data with `!is.na(data$iron)` to remove missing values. Created scatter plot with `geom_point()` where bubble size and color represent iron content, using `scale_size_continuous()` and `scale_color_gradient()` (yellow=low iron, red=high iron).

Discussion

Chart reveals moderate positive correlation between carbs and calories. Main clusters: (1) Low-carb, low-calorie (0-40g carbs, 0-500 kcal) - most foods with low iron; (2) Middle region (10-50g carbs, 200-700 kcal) - one prominent large red bubble at ~55-60g carbs, ~250 kcal indicating high iron (7-10mg); (3) High-calorie outliers (~1450 kcal) at both low and high carbs - yellow bubbles showing low iron despite high energy.

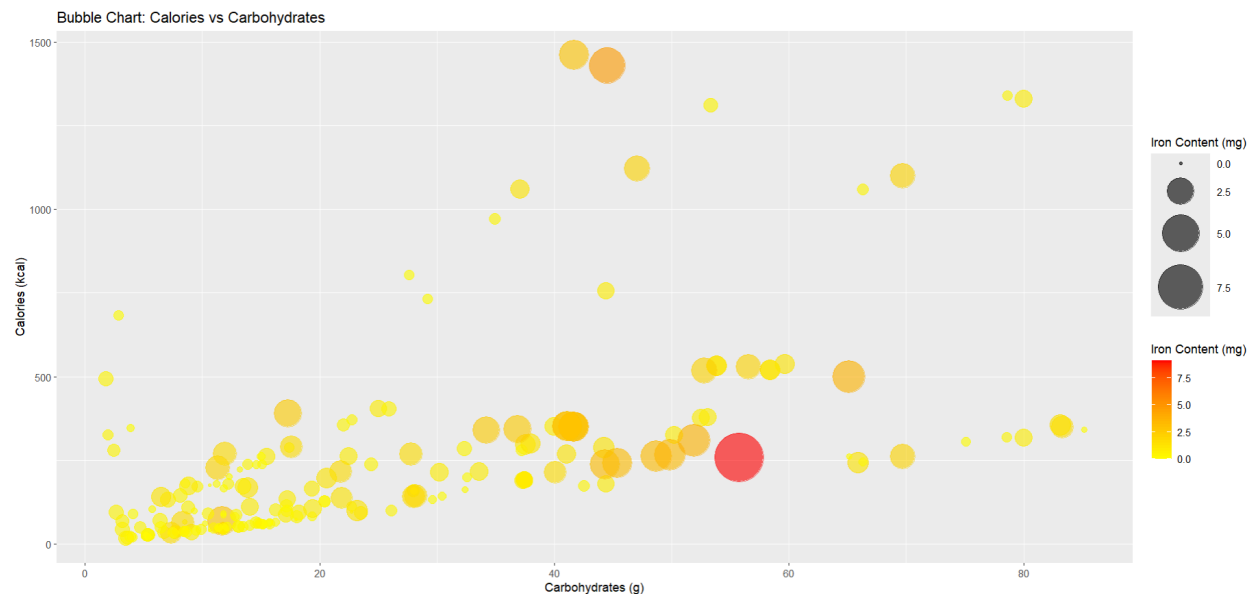


Figure 15.1: Bubble chart showing calories vs carbohydrates, bubble size/color = iron content

Iron content is independent of calories and carbohydrates - high-iron foods appear across limited ranges while low-iron foods distribute widely. The large red bubble likely represents fortified cereals or legumes - optimal for iron supplementation with moderate calories. High-calorie outliers show low iron despite high energy, representing "empty calories." For iron deficiency, target the middle region with larger red or orange bubbles. For weight management, focus on low-calorie region regardless of iron. The multi-dimensional visualization reveals that achieving balanced nutrition requires strategic food selection across different nutritional profiles.

Question 16: Nutrient Ratio Analysis

a. Calculate the protein-to-calorie ratio for each food item.

Logic

1. We created a new column called 'ratio' in the dataframe
2. A pipeline is used and the **mutate()** function appends the new ratio column to the existing dataset while preserving the structure
3. Inside the **mutate()**, an **ifelse()** statement is used to insert a ratio value to each row. This is calculated by doing protein/calories.
4. The **ifelse()** statement will also check if the calories being divided is greater than zero to avoid dividing by zero errors. If it is, the division will occur. If not, the ratio is set to 0.

Discussion

This task focuses on feature engineering to create a new variable that makes the analysis more effective. The **ifelse()** statement also protects against any zero calorie entries in the dataset by setting the ratio to zero.

b. Identify the top 10 foods with the highest ratio

Logic

1. **arrange()** function is used to sort the ratio values in descending order, with the highest ratio first.
2. **slice_head(n=10)** is applied to only show the first 10 values.
3. Food name is printed alongside the ratio.

Discussion

```
[1] "Top 10 Foods by Protein-to-Calorie Ratio:"
> print(top_10_foods %>% select(food_name, ratio))
      food_name      ratio
1  Pork with chili and tomatoes 0.12252632
2  Yogurt, Greek, Blueberry, CHOBANI 0.08817073
3  Yogurt, Greek, 2% fat, mango, CHOBANI 0.08215054
4  Grape leaves, canned 0.06188406
5  Pepper, banana, raw 0.06148148
6  Peppers, banana, raw 0.06148148
7  Onions, green, raw 0.05718750
8  Orange chicken 0.05519084
9  Sun-dried tomatoes 0.05468992
10 Kefir, lowfat, strawberry, LIFEWAY 0.05467742
```

Figure 16.1: Output of top 10 foods based on protein-to-calorie ratio

This task uses filtering and ranking data to identify the top values, in this case the top 10 ratios of protein to calories that were identified in part (a). Combining **arrange()** and **slice_head()** is better and cleaner than using array indices, as **slice_head** handles grouped data better, rather than with arrays. The results here successfully show the top 10 food items with 'Pork with chilli and tomatoes' with the highest ratio of 0.123. The next four items contain Greek yoghurt, grape leaves, and peppers, with orange chicken coming in at the 8th position. This indicates that while dairy and meat products have the highest protein content and make up the majority of this top 10, there are non-meat and dairy products that challenge this assumption.

c. Visualize these top foods using a horizontal bar chart.

Logic

1. A **ggplot** object is created using the filtered 'top_10_foods' dataframe.
2. The x-axis maps the food_name which is reordered by ratio value to ensure the bars are sorted by length.
3. The **geom_col()** renders the bar chart.
4. **coord_flip()** rotates the chart 90 degrees.

Discussion

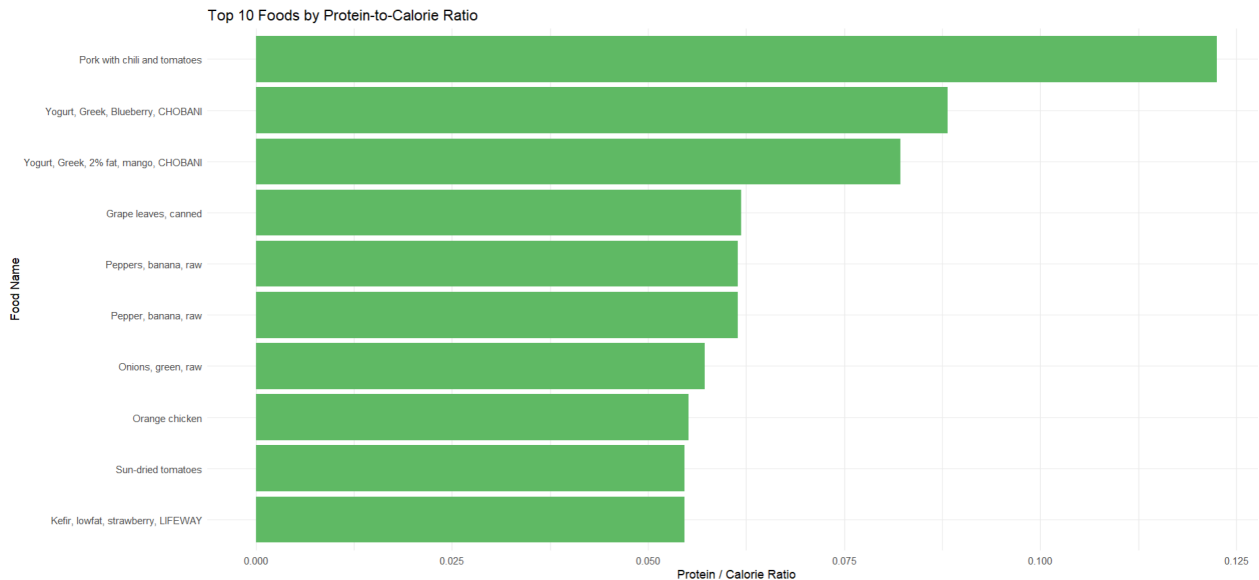


Figure 16.2: Bar chart showing top 10 foods based on protein-to-calorie ratio

This task tests our ability to communicate the ranked data from (b) in a visual format. The results clearly shows the gap between the first placed item based on ratio and the second-placed item. The visualisation makes it much easier to see how large this gap is, and allows viewers to spot patterns of repeated entries. Using a protein-to-calorie ratio normalizes protein intake by energy density, making it more informative for identifying nutritionally efficient foods than absolute protein values alone.

Question 17: Macronutrient Contribution

- Compute percentage contribution of protein, carbs, and fat to total calories.
- Create a stacked bar chart showing macronutrient composition by category.

Logic

We used the `aggregate()` function to sum the total grams of protein, carbs, and fat for each category. Then, we applied the standard Atwater factors (4 kcal/g for protein and carbs, 9 kcal/g for fat) to convert these masses into calories, which align with nutritional standards. We calculated the percentage contribution of each nutrient to the total calories. Finally, the data was reshaped and plotted as a stacked bar chart using `geom_bar(position = "stack")`. A normalised stacked bar chart was used to make sure the categories sum to 100% to allow for more direct comparison

Discussion

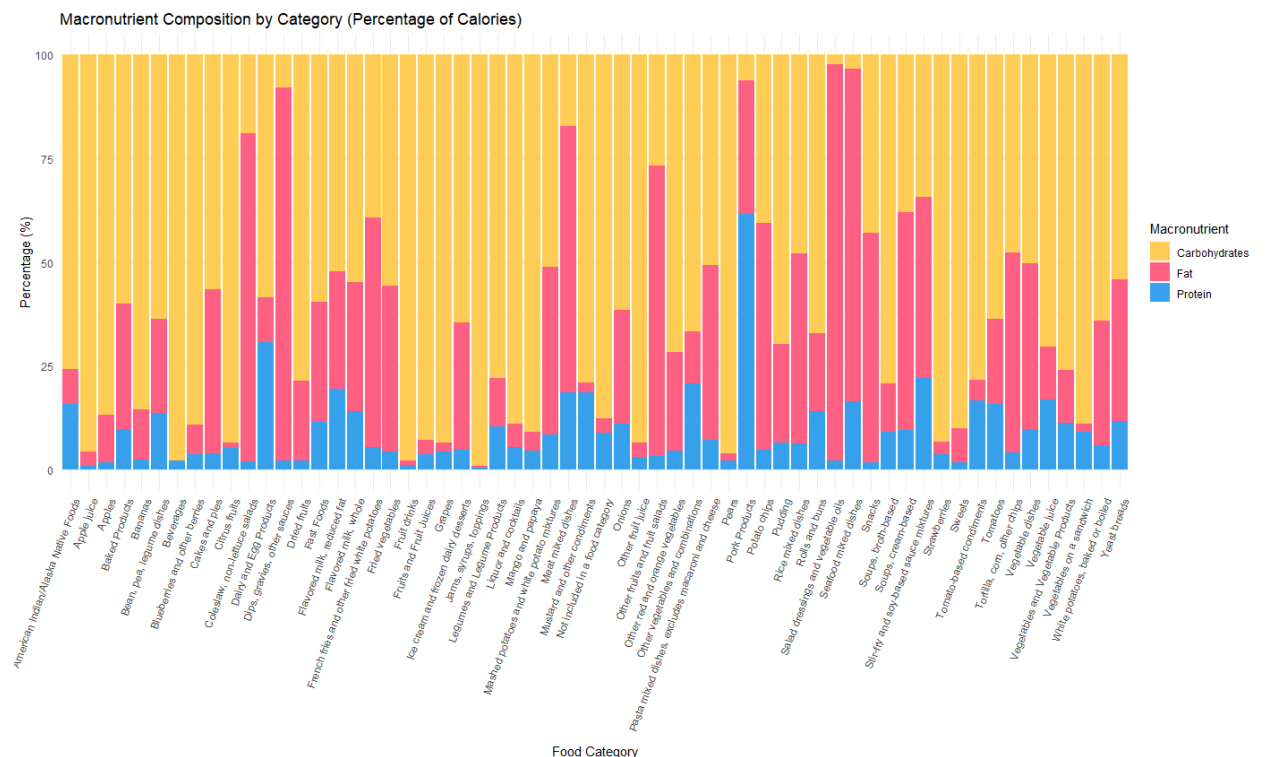


Figure 17.1: Stacked bar chart displaying macro nutrient composition by category

This visualization is critical for understanding the "nutritional fingerprint" of each food category.

- **Composition:** The chart uses color to show where the calories come from. As expected, the 'Meats' category is dominated by **Protein (Blue)** and **Fat (Red)**, with very little Carbohydrate.
- **Contrast:** In contrast, categories like 'Grains' or 'Fruits' show a dominant yellow section, representing **Carbohydrates**.
- **Application:** This chart helps to quickly identify which food categories align with specific dietary goals, such as high-protein or low-carb diets.

